EME: Energy-Based Multiexpert Model for Long-Tailed Remote Sensing Image Classification

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***Abstract*— The distribution of remote sensing scene images often follows a long-tailed pattern, where there is an abundance of samples in a few dominant classes and a scarcity of samples in most other classes. This presents two major challenges when it comes to identifying this type of data: *Head-Dominance:* Models trained on such data tend to prioritize the dominant classes, overlooking the tail classes and resulting in poor performance when it comes to recognizing them. *Tail-Interference:* The presence of tail classes disrupts the learned representations for the head classes, acting as noise that negatively impacts the recognition accuracy of the head data. To address these challenges, we propose an innovative solution called the energy-based multiexpert (EME) model. The core concept behind this approach is to utilize energy-based discriminators (EDors) to separate the data into head and tail categories. Subsequently, we design multiple experts to classify the head and tail data separately, ensuring that the significant differences in data volume between these categories do not interfere with each other. Experimental results obtained by applying the EME model to three remote sensing datasets demonstrate its efficiency, outperforming current state-of-the-art (SOTA) methods. These findings underscore the effectiveness of our proposed approach in addressing the challenges posed by the long-tailed distribution in remote sensing scene images.**

***Index Terms*— Energy-based discriminators (EDors), head-dominance, long-tailed distribution, remote sensing image classification, tail-interference.**

I. Introduction

In RECENT years, deep convolutional neural networks (CNNs) have achieved remarkable success in various computer vision tasks [1], [2], [3], [4], [5], [6]. However, the predominant training of models on large curated datasets with relatively balanced class distributions presents a challenge when applied to real-world remote sensing scene images [7], [8], [9], [10], [11]. Because these types of images often exhibit a long-tailed distribution, where only a few classes have a substantial number of samples, while the majority of classes have limited samples [see Fig. 1(a)].

After conducting thorough research and analysis, this article identifies two key reasons why general models struggle to handle long-tailed problems. 1) *Head-Dominance*: This occurs when head classes with an abundance of samples dominate the training process, leading to poorer performance on tail classes [Fig. 1(b)]. 2) *Tail-Interference*: This refers to the presence of tail classes acting as noise, negatively impacting the optimal classification of head classes [Fig. 1(c)]. To evaluate our assumptions, we conducted experiments depicted in Section V-G. These factors underline the complexity associated with effectively training models on long-tailed distributed data and emphasize the need to address the limitations posed by imbalanced class distributions. It is crucial to develop techniques that can mitigate the negative impacts of head-dominance and tail-interference. This allows us to achieve better performance and improve the generalisability of models when dealing with long-tailed data distributions.

The current research methods for addressing long-tailed problems have shown some progress. They can be broadly categorized into two approaches: class re-sampling/re-weighting and ensemble learning. However, these methods do not adequately consider the aforementioned factors. Specifically

1) Class re-sampling approaches [12], [13], [14], [15] involve over-sampling tail classes and under-sampling head classes to balance the class distribution. Meanwhile, re-weighting methods [16], [17], [18], [19] assign higher loss weights to tail classes to prioritize their learning. Although these methods effectively influence the update of classifier weights in deep networks and promote learning, they risk over-focusing on tail classes at the expense of accuracy on head classes.

2) Ensemble learning methods [20], [21], [22] train separate models for different data groups and then combine their predictions. However, apart from potentially suboptimal ensemble strategies, interference between head and tail data during joint model training may also lead to a decline in performance for the head class. Therefore, advanced technology is needed to address the challenges posed by head-dominance and tail-interference in long-tailed problems effectively.

Considering the current research status and existing problems, we propose an energy-based multiexpert (EME) model for remote sensing scene image classification (see Fig. 2). Our

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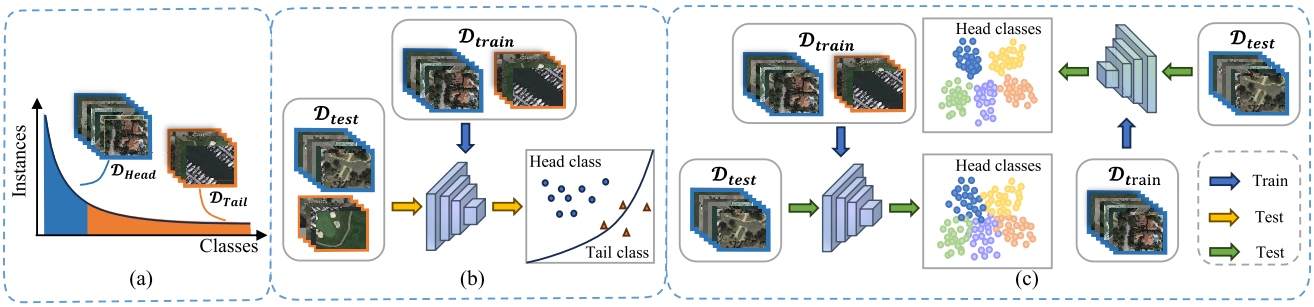


Fig. 1. Illustration of the mutual influence between head and tail classes arising from the distribution imbalance in long-tailed data. (a) *Long-Tailed Distribution*: A few head classes have abundant samples while most tail classes have limited samples. (b) *Head-Dominance*: A model trained on imbalanced data favors head classes, leading to misclassification of some tail samples as head (c) *Tail-Interference*: When classifying head data, training the model exclusively with head data produces more distinct class boundaries compared to training the model with a combination of head and tail data.

core idea is to address the interference between head and tail data by implementing a hierarchical approach that processes these two types of data separately. We handle the head and tail subclasses individually, ensuring that their unique characteristics are accurately captured during the classification process. Additionally, through experiments in Section V-D and surveys [7], [21], [22], we have recognized the significance of the middle class in remote sensing scene image classification. This class holds valuable information that can substantially enhance the overall accuracy of the classification system. Therefore, we integrate the middle class into our approach, providing it with equal consideration alongside the head and tail classes. Our EME model consists of two major components.

1) *Coarse-Grained Discriminator (CGD)* Block: Building upon the inspiration from [23], we incorporate an energy-based loss function for effectively discriminating the head, middle, and tail classes. This block comprises two energy-based discriminators (EDors). The first primarily focuses on classifying the data into the head subclass and other classes, while the second handles data belonging to the tail subclass and other classes. By considering the decisions from both discriminators, we achieve effective classification across the head, middle, and tail subclasses. (2) *Multiexpert Learning (MEL)* Block: In this component, we train three expert models based on the classification results derived from the CGD block. Each expert model is specifically trained to handle one subclass: head, middle, or tail. During the inference stage, our approach incorporates a decision-interaction (DI) strategy to enhance interactions among multiple experts. This strategy aims to address any potential errors that might occur due to the limitations of the CGD block. By promoting collaborative decision-making, we leverage the unique strengths of each expert, thereby improving the overall classification performance. The important abbreviations and notations used in this work are defined in Table I.

In summary, our contributions are summarized as follows.

1) We emphasize that a fundamental challenge in addressing the long-tailed problem is the interference between the head and tail classes. To overcome this obstacle, we propose the EME model, which effectively blocks the interference between these two class subsets, subsequently leading to significant improvements in model performance.
2) We devise two complementary blocks: CGD and MEL. These blocks are designed to work in tandem and collaborate with each other, allowing us to achieve our desired goals effectively.
3) We conducted extensive experiments on three widely used long-tailed remote sensing datasets to evaluate the performance of our proposed approach. The results demonstrated significant improvements compared to other SOTA methods, validating the effectiveness and superiority of our approach.

TABLE I
IMPORTANT ABBREVIATIONS AND NOTATIONS

Abbreviation and notation	Definition
CGD	coarse-grained discriminator
MEL-C	coupled multi-expert learning
MEL-D	decoupled multi-expert learning
EDors	energy-based discriminators
DI	decision-interaction
ID	in-distribution
OOD	out-of-distribution
$\mathcal{D}_H$, $\mathcal{D}_M$, $\mathcal{D}_T$	head data, middle data, tail data
$\mathcal{D}_{id}$, $\mathcal{D}_{ood}$	in-distribution data, out-of-distribution data
L_c	cross-entropy classification loss
L_e	energy loss
$f(\cdot)$	CNN-based classifier for discriminators
$\varphi(\cdot)$	CNN-based classifier for experts

II. Related Work

A. Long-Tailed Recognition

Currently, methods to address the long-tailed problem can be broadly categorized into two approaches: class re-sampling/re-weighting and ensemble learning.

1) Class re-sampling methods [12], [13], [14], [15] involve over-sampling tail classes and under-sampling head classes to balance the class distribution. For example, SMOTE [13], a classical over-sampling method, generates new minority samples by mixing several intra-class adjacent samples. Inverse random under-sampling [15] severely undersamples majority classes. Meanwhile, re-weighting methods [16], [17], [18], [19] assign higher loss weights to tail classes to prioritize their learning. For instance, class balanced (CB) loss [17] uses effective class numbers to approximate expected sample sizes and reweights terms inversely to address imbalance. Focal loss [18] reweights classes based on prediction probabilities to assign higher weights to harder tail classes. Although effective at influencing classifier updates, these methods risk over-focusing on tails at the expense of head accuracy.

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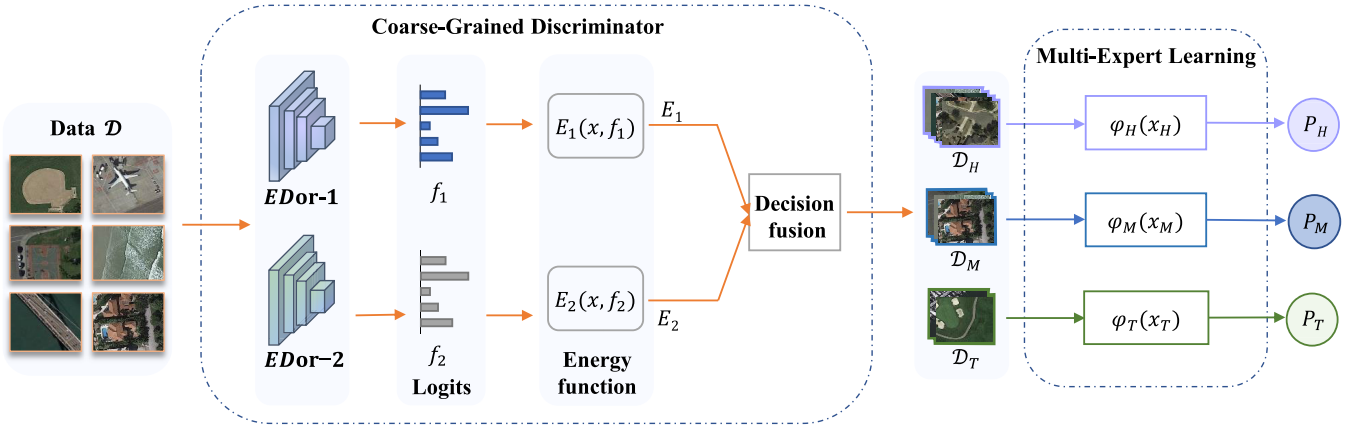


Fig. 2. Framework of the proposed EME Model. The EME model comprises two blocks: a CGD block and a MEL block. The input sample x is first fed into two independent energy-based discriminators, EDor-1 and EDor-2 (see Fig. 3 for EDors details), which output logits f_1 and f_2. The energy functions then calculate energies E_1 and E_2 from f_1 and f_2. E_1 and E_2 are fused to determine whether sample x belongs to the head, middle, or tail category (see Fig. 4 for decision fusion details). Finally, the sample x is routed to the appropriate expert, φ_H, φ_M or φ_T, for classification.

2) Ensemble methods [7], [20], [21], [22], [24] train separate models on different data splits and combine predictions. For example, BBN [20] uses two network branches: one with uniform sampling simulating the original distribution, the other with reverse sampling for more tail examples. In the training process, the prediction of the two branches is dynamically combined, gradually shifting the learning focus from head-to-tail classes. T2FTS [7] divides classes into relatively balanced subclasses with separate teacher models and then distills knowledge into a student. However, these integration methods train all experts simultaneously on the complete dataset when aggregated. This can lead to mutual interference between the head and tail classes, reducing the performance of the head classes. Our approach introduces a CGD block before the expert models. By separating the head, medium, and tail subsets early in the process, we mitigate the interference issue. Using an energy-based model, the discriminator can efficiently differentiate subgroups of data, providing stronger isolation between components than previous integration methods.

B. Long-Tailed Problem in Remote Sensing

In real-world remote sensing applications, long-tailed data distributions are very common. This is not only caused by the high labeling costs but also because different land cover types have vastly different levels of spatial coverage. This results in some classes being well-represented while others are limited in training datasets [25]. Such data imbalance leads to poor recognition performance of tail classes using standard deep learning schemes. To alleviate the long-tailed problem in remote sensing image classification, some methods have been proposed: Douzas et al. [26] used the Geometric-SMOTE oversampling technique to improve prediction accuracy for minority tail classes. Feng et al. [10] proposed a semi-supervised long-tailed learning framework based on spatial neighborhood information (SLN-SNI) for hyperspectral image classification. Zhao et al. [7] divided the data into three relatively balanced subgroups, trained teacher models for each, and then refined a student model for long-tailed classification. Liu et al. [8] proposed a loss reweighting method that constructs class weights based on cumulative classification scores to address the long-tailed data distribution problem in remote sensing image classification. However, existing methods have achieved limited improvements, and the long-tailed distribution remains a key challenge in remote sensing image analysis. More advanced techniques are needed to effectively mitigate the data imbalance issue.

C. Energy-Based Learning

Energy-based learning aims to model data distributions by defining an energy function, which assigns low energy values to probable data points and high energy values to improbable ones [27]. Energy-based learning methods have shown promising performance on various machine learning tasks. Early energy-based models such as Boltzmann machines [28] aim to estimate data distribution by minimizing energy functions. Recent work has incorporated deep neural networks into the energy function modeling [29], [30], [31], [32], leading to more powerful feature learning capacities. For example, the deep structured energy model (DSEBM) [30] combines energy scores and data errors for anomaly detection. Energy-based constrained decoding with langevin dynamics (COLD) [31] applies the energy function to different text generation tasks, including lexical constraint generation, inductive reasoning, and counterfactual reasoning. Du and Mordatch [32] show that EBMs are useful across a wide variety of tasks, achieving SOTA out-of-distribution (OOD) classification, adversarially robust classification, SOTA continual online class learning, and coherent long-term predicted trajectory rollouts. Our work follows the discriminative modeling principle of energy-based learning. We design energy functions to measure sample similarity within subgroups. By optimizing cluster-specific energy scores, the model learns to effectively differentiate data clusters like head and tail.

III. Problem Setup

Long-tailed recognition involves learning from a training dataset that exhibits a long-tailed distribution. In such a distribution, the majority of classes are represented by only a few

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samples, whereas a small number of classes have a significant abundance of samples (see Fig. 1). Given training set $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}$, where $y_i \in \{1, 2, \ldots, C\}$ is the corresponding label for each sample x_i. $\sum_1^C n^c = N$, where n^c is the number of samples of the c_{th} class. The imbalance ratio $\alpha = n^c_{\min}/n^c_{\max}$ measures how imbalanced the long-tailed training set is. If $\alpha \leq 1$, it emphasizes classification accuracy averaged over classes, i.e., accuracy $= (1/C)\sum_1^C \text{acc}_c$ is used as the main metric, where acc_c is the top-1 accuracy of class-c.

IV. Methodology

A. Overview

This article presents the EME Model (see Fig. 2) for long-tailed remote sensing image classification. The model comprises two primary components.

1) *CGD Block:* The objective of this block is to partition the input data into multiple independent subsets that are not mutually influenced. In line with experimental findings (Table IV) and expertise [7], [21], [22], the data is divided into three distinct subcategories: head, middle, and tail. This article adopts energy-based models to achieve this partition.
   Specifically, this block consists of two EDors. One EDor is dedicated to classifying data as either the head or middle-tail category, while the other focuses on classifying data as either the tail or head-middle category. During inference, a voting process is applied to the category decisions from the two EDors. The data is classified into the category receiving the most votes. Additionally, a decision fusion strategy is introduced to address potential conflicts in the classification process. This effectively partitions the data into distinct head, middle and tail classes.
2) *MEL Block:* Based on the head, middle, and tail data obtained from the CGD block, three specialized classifiers are designed for the subcategories. This design enables the model to mitigate the effects of "Head-dominance" and "Tail-interference," thus ensuring that these factors have minimal impact on the overall classification performance.
   In the specific implementation process, two schemes are designed: Coupled MEL (MEL-C) and Decoupled MEL (MEL-D). MEL-C indicates that our method is an end-to-end network where CGD and MEL are closely interconnected. The two blocks are simultaneously updated based on the final results. MEL-D, on the other hand, implies that the CGD and MEL modules operate independently without influencing each other. MEL-D offers advantages in terms of time and effort savings compared to MEL-C. However, it introduces new drawbacks. In this case, the CGD module may generate errors when predicting the head, middle, and tail categories. Since the CGD module cannot be corrected by the final classification results from MEL, these errors can persist and affect the overall outcome. To minimize these effects, this article maintains the approach of independently training the head, middle, and tail experts during the training phase. However, in the inference phase, a DI strategy is introduced. Test images are individually processed through three frozen expert models, and the image category is determined based on the final prediction results obtained from the experts.

B. Coarse-Grained Discriminator

1) Review of Energy-Based Model: The essence of the energy-based model [27] is to construct an energy function $E(x) = \mathbb{R}^K \to \mathbb{R}$, where K is the dimension of the input space. The energy function can map each input x in the input space to a scalar, called the energy. The set of energy values can be converted into a probability density $p(x)$ through a Boltzmann distribution

$$p(y \mid x) = \frac{e^{-E(x,y)/T}}{\int_{y'} e^{-E(x,y')/T}} = \frac{e^{-E(x,y)/T}}{e^{-E(x)/T}} \tag{1}$$

where T is the temperature parameter and the denominator $\int_{y'} e^{-E(x,y')/T}$ is the partition function that normalizes the distribution. The Helmholtz free energy $E(x)$ at point $x \in \mathbb{R}^K$ can be expressed as the negative logarithm of the partition function

$$E(x) = -T \cdot log \int_{y'} e^{-E(x,y')/T}. \tag{2}$$

By designing a suitable energy function $E(x, y)$, the model can assign low energy to "realistic" or "high probability" data samples and high energy to "unrealistic" or "low probability" data samples. This method enables the identification of in-distribution (ID) data and OOD data [23].

2) EDors for Long-Tailed Recognition: Drawing inspiration from the seminal works of [27] and [23], we introduce a novel methodology that leverages two independent EDors to effectively categorize data into head, middle, and tail segments. In the initial step, we designate the training head data as in-distribution, while considering all other data as OOD. By employing this configuration, we train an EDor-1 model specifically tailored for this purpose. Subsequently, we assign the training tail data as in-distribution, with the remaining data categorized as OOD. This setup allows us to train a dedicated EDor-2 model to accurately classify the data. Finally, by utilizing the logits obtained from both EDors, we are able to determine the appropriate categorization of the data, differentiating between head, middle, and tail segments. The following parts provide a comprehensive elucidation of the entire process involved in our proposed methodology.

Consider the neural network classifier $f(x) = \mathbb{R}^K \to \mathbb{R}^C$, which maps the sample x to C logits. These logits derive the probability distribution of the category through the softmax function

$$p(y \mid x) = \frac{e^{f_y(x)/T}}{\sum_1^C e^{f_c(x)/T}} \tag{3}$$

where $f_y(x)$ represents the logit of the y_{th} class lable. By relating (1) and (3), the energy of a given input (x, y) is defined as $E(x, y) = -f_y(x)$. The free energy function $E(x; f)$ of

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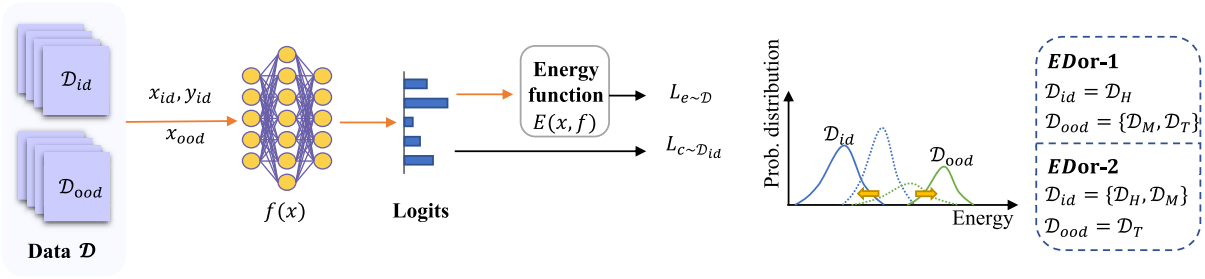


Fig. 3. Training of EDors EDor-1 and EDor-2. In-distribution samples (x_{in}, y_{in}) from $\mathcal{D}_{\text{id}}$ and OOD samples x_{ood} from $\mathcal{D}_{\text{ood}}$ are input into the CNN model $f(x)$, which classifies all samples into D_{id} categories and outputs corresponding logits. The energy function $E(x, f)$ then calculates sample energies based on logits. The model is trained to maximize the energy difference between $\mathcal{D}_{\text{id}}$ and D_{ood}, based on the energy loss $L_{e\sim\mathcal{D}}$ on $\{\mathcal{D}_{\text{id}}, \mathcal{D}_{\text{ood}}\}$ and the classification loss $L_{c\sim D_{\text{in}}}$ on D_{in}. For EDor-1, $\mathcal{D}_{\text{id}}$ is the head data $\mathcal{D}_H$, and $\mathcal{D}_{\text{ood}}$ is the combined set of middle and tail data $\{\mathcal{D}_M, \mathcal{D}_T\}$. For EDor-2, $\mathcal{D}_{\text{id}}$ is the combined set of head and middle data $\{\mathcal{D}_H, \mathcal{D}_M\}$, and $\mathcal{D}_{\text{ood}}$ is the tail data $\mathcal{D}_T$.

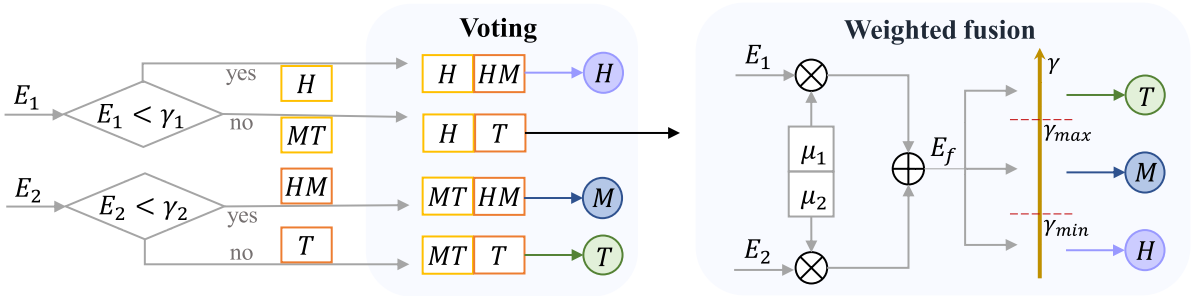


Fig. 4. Overview of decision fusion. Energy E_1 from EDor-1 is first compared to γ_1. If $E_1 < \gamma_1$, the sample belongs to the head category; otherwise, it belongs to the middle or tail category. Similarly, E_2 from EDor-2 is compared to γ_2. If $E_2 < \gamma_2$, the sample belongs to the head or middle category; otherwise, it belongs to the tail category. The results are then voted on, and the sample belongs to the category with the most votes. For ambiguous cases, E_1 and E_2 are weighted by μ_1 and μ_2, respectively, and summed to obtain E_f. The relationship between E_f, γ_{min} and γ_{max} then determines if the sample belongs to the head, middle or tail, where $\gamma_{\text{min}} = min\{\gamma_1, \gamma_2\}$, $\gamma_{\text{max}} = max\{\gamma_1, \gamma_2\}$.

the neural network $f(x)$ over $x \in \mathbb{R}^K$ can be expressed as

$$E(x; f) = -T \cdot \log \sum_{1}^{C} e^{f_c(x)/T}. \tag{4}$$

The energy function calculates the energy of each input sample x. The energy of a sample with a high probability is lower than that of a sample with a low probability. Therefore, it can be used to determine whether a sample is OOD or not. By fine-tuning the neural network, lower energy is allocated to ID data, and higher energy is allocated to OOD data so that ID and OOD data are more distinguishable. The objective function is defined as follows:

$$\min_{\theta} \quad L_{c\sim\mathcal{D}_{\text{id}}} + L_{e\sim\mathcal{D}} \tag{5}$$

where $L_{c\sim\mathcal{D}_{\text{id}}}$ represents the cross-entropy classification loss of the ID training data $\mathcal{D}_{\text{id}}$. The energy loss $L_{e\sim\mathcal{D}}$ consists of the energy loss of the ID and OOD data:

$$\begin{aligned} L_{e\sim D} = &(\max\{0, E(x_{\text{id}}) - m_{\text{id}}\})^2 \\ &+ (\max\{0, m_{\text{ood}} - E(x_{\text{ood}})\})^2 \end{aligned} \tag{6}$$

where $m_{\text{id}} < m_{\text{ood}}$ are two independent margin hyperparameters. The model penalizes ID samples x_{id} whose energy is higher than the margin parameter m_{id} while penalizing OOD samples x_{ood} whose energy is lower than the margin parameter m_{ood}.

During inference, the energy is interpreted as the OOD score, and OOD detection is conducted by using (4).

$$g(x; \gamma, f) = \begin{cases} 1, & \text{if } E(x, f) < \gamma \\ 0, & \text{if } E(x, f) \geq \gamma \end{cases} \tag{7}$$

where γ denotes the energy threshold. OOD detector $g(x)$ classifies inputs with energy below this threshold as ID category and inputs above this threshold as OOD category.

To effectively utilize specialized models for different data subgroups and minimize interference, we employ a two-step approach to partition the full dataset $\mathcal{D}$ into head data $\mathcal{D}_H$, middle data $\mathcal{D}_M$, and tail data $\mathcal{D}_T$ using two independent EDors. Each segmented subset is then assigned to its respective expert model.

For EDor-1, the head data $\mathcal{D}_H$ is treated as ID, while the middle data $\mathcal{D}_M$ and tail data $\mathcal{D}_T$ are considered OOD. The objective in (5) seeks to widen the energy gap between $\mathcal{D}_H$ and the combined OOD set $\{\mathcal{D}_M, \mathcal{D}_T\}$. During inference, the threshold γ_1 on energy E_1 determines whether input x belongs to the head or middle-tail category

$$x \in \begin{cases} \mathcal{D}_H, & \text{if } E_1 < \gamma_1 \\ \mathcal{D}_M \text{ or } \mathcal{D}_T, & \text{if } E_1 \geq \gamma_1. \end{cases} \tag{8}$$

For EDor-2, the head data $\mathcal{D}_H$ and middle data $\mathcal{D}_M$ are ID, while the tail data $\mathcal{D}_T$ is OOD. Similarly, the objective in (5) aims to increase the energy gap, this time between the combined ID set $\{\mathcal{D}_H, \mathcal{D}_M\}$ and the OOD tail set $\mathcal{D}_T$. The inference threshold γ_2 applied on E_2 categorizes x into

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head-middle or tail

$$x \in \begin{cases} \mathcal{D}_H \text{ or } \mathcal{D}_M, & \text{if } E_2 < \gamma_2 \\ \mathcal{D}_T, & \text{if } E_2 \geq \gamma_2. \end{cases} \tag{9}$$

The decision fusion module in the CGD block is illustrated in Fig. 4. The input x is passed through EDor-1 and EDor-2 to obtain the energies E_1 and E_2, respectively. Subsequently, (8) and (9) compare E_1 and E_2 with the thresholds γ_1 and γ_2, resulting in two separate decision outcomes. These outcomes are then subjected to a voting process, where the sample is assigned to the category that receives the majority of votes. In cases of ambiguity, such as voting ties, the energies E_1 and E_2 are weighted and combined into a fused energy E_f

$$E_f = \mu_1 E_1 + \mu_2 E_2 \tag{10}$$

where $\mu_1 + \mu_2 = 1$, are the weight parameters. E_f less than $\gamma_{\min}$ indicates the head, greater than $\gamma_{\max}$ indicates the tail, and between $\gamma_{\min}$ and $\gamma_{\max}$ indicates the middle. where $\gamma_{\min} = \min\{\gamma_1, \gamma_2\}$, $\gamma_{\max} = \max\{\gamma_1, \gamma_2\}$.

C. Multiexpert Learning

To alleviate the detrimental impact of long-tailed distributions on prediction accuracy, we propose a MEL block. The dataset $\mathcal{D}$ is partitioned into head, middle, and tail subsets ($\mathcal{D}_H$, $\mathcal{D}_M$, and $\mathcal{D}_T$). We proceed to train three independent experts (φ_H, φ_M and φ_T) specifically for the head, middle, and tail data subsets. In addition, depending on whether MEL is updated in conjunction with CGD, the MEL block can be classified into two modes: coupled and decoupled.

The Coupled MEL (MEL-C) block is an end-to-end architecture that updates the network parameters in coordination with the CGD block. This approach offers the advantage of improved collaboration between the two blocks. However, a drawback is that training them together consumes significant resources, and adjusting parameters becomes challenging. Consequently, reaching the theoretical upper bound can be difficult to achieve.

The decoupled MEL and CGD blocks operate independently without any mutual influence. It is important to highlight that, for enhanced efficiency, we utilize the pre-trained EDor-1/EDor-2 as the initial pre-trained model of the multiple experts. Subsequently, we make parameter adjustments based on this foundation. In the inference phase, to mitigate any classification errors that may arise due to the limitations of the CGD block, we facilitate DI strategy between multiple experts. Assuming $x_{\tilde{H}}$ is a sample assigned to the head group by the CGD block, where $\tilde{H}$ indicates that the true class of the sample may not be head. The sample is then classified separately by the three expert models

$$p_{\tilde{H}-h} = \max_h\{\text{softmax}(\varphi_H(x_{\tilde{H}}))\} \tag{11}$$

$$p_{\tilde{H}-m} = \max_m\{\text{softmax}(\varphi_M(x_{\tilde{H}}))\} \tag{12}$$

$$p_{\tilde{H}-t} = \max_t\{\text{softmax}(\varphi_T(x_{\tilde{H}}))\} \tag{13}$$

where $p_{\tilde{H}-h}$, $p_{\tilde{H}-m}$, $p_{\tilde{H}-t}$ represent the probabilities of the coarse-grained head sample $x_{\tilde{H}}$ belonging to classes h, m, and t, as predicted by the head, middle, and tail experts respectively. Normally, sample $x_{\tilde{H}}$ would be considered belonging to class h. However, the CGD block may make incorrect predictions. To mitigate potential errors, we apply the following rule:

If $p_{\tilde{H}-h} < 0.5$ and either $p_{\tilde{H}-m}$ or $p_{\tilde{H}-t}$ is greater than 0.9, the sample $x_{\tilde{H}}$ is revised to be classified into the class with the highest probability between m and t. Otherwise, the sample $x_{\tilde{H}}$ is determined to belong to class h as initially predicted.

By partitioning the data into manageable subsets and training dedicated experts on each subset, we enable each expert model to focus on characteristics pertinent to its associated data partition. This mitigates the challenges of learning from the imbalanced long-tailed data distribution. EDor-1/EDor-2 provides a strong initialization for the expert models by extracting a common set of features from the input. The resulting multiexpert system benefits from divide-and-conquer learning on manageable data subsets and ensemble effects, outperforming a single model facing the full complexity of the imbalanced long-tailed distribution.

In summary, the EME model provides an effective framework to address the long-tailed distribution problem. It utilizes EDors to efficiently separate the head, middle and tail data, and then designs expert models to deal with different levels of data respectively. This separation and specialized modeling improve classification performance under imbalanced data settings. This makes the EME model potentially suitable for a wide variety of tasks involving long-tailed or class imbalance issues, such as scene graph generation [33], [34], object detection [35], [36], and medical image analysis [37], [38].

V. Experiments

A. Datasets

DOTA [44] exhibits an imbalanced long-tailed distribution with 15 classes and 188 282 instances in total. We select 98 906 images for training and 28 853 images for testing. Each class has 325 to 28 068 images, and the imbalance ratio is 0.01.

FGSC-23 [45] is a long-tailed optical remote sensing dataset for fine-grained ship target identification. It contains a total of 4052 ship image samples with 23 types of ship subclasses. We use 3256 images for training and the remaining 825 for testing. The number of training images per class ranges from 17 to 434, with an imbalance ratio of 0.04.

PatternNet [46] contains 38 classes of remote sensing scenes, with originally 800 images per class. To emulate the long-tailed distribution of real-world data, we first split each class into 100 images for test set and the remaining images for training set. Then for the training set, we apply an exponential decay function sampling

$$n = n^c \alpha^{\frac{c}{C-1}} \tag{14}$$

where c is the class index, n^c is the original number of samples for class-c in the training set, α is the imbalance ratio, and C is the total number of classes. We analyze the performance under four different imbalance ratios of $\alpha = 0.1, 0.05, 0.02$ and 0.01.

Considering the class distribution of each dataset, we partition them into three balanced subgroups: head, middle, and tail. The distributions of samples for these three datasets are displayed by the red line in Fig. 5.

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TABLE II

PERFORMANCE COMPARISON OF THE PROPOSED EME MODEL AND OTHER METHODS ON PATTERNNET, DOTA, AND FGSC-23 DATASETS. THE TABLE LISTS AVERAGE TOP-1 ACCURACY (%) ON HEAD, MIDDLE, AND TAIL CLASSES AS WELL AS OVERALL ACCURACY (%). NOTE THAT OVERALL ACCURACY IS CONSIDERED THE MOST IMPORTANT METRIC, WHILE THE REMAINING RESULTS ARE PROVIDED FOR REFERENCE. TO EMPHASIZE THIS, WE HAVE SHADED THE OTHER RESULTS

Method	PatternNet				DOTA				FGSC-23			
	Head	Middle	Tail	All	Head	Middle	Tail	All	Head	Middle	Tail	All
LDAM [39]	90.38	76.54	78.75	81.97	57.29	91.62	60.49	72.30	56.99	62.59	56.38	58.13
BKD [40]	90.15	72.46	78.25	80.34	85.97	92.52	81.95	86.98	63.71	55.18	65.46	62.40
SADE [41]	92.38	74.46	82.75	83.21	57.40	94.33	87.97	84.40	51.55	61.73	70.52	64.11
T2FTS [7]	94.08	87.69	88.75	90.21	78.26	91.31	87.80	87.29	71.20	**79.46**	71.46	73.49
BBN [20]	87.38	74.23	76.50	79.45	38.93	**94.95**	83.38	80.93	21.83	24.73	50.64	37.62
RIDE [22]	92.15	75.00	83.25	83.47	64.98	94.00	75.14	80.65	45.15	46.48	44.53	45.17
NCL [42]	95.00	84.46	89.08	89.53	68.86	82.03	63.48	71.97	36.24	47.37	51.90	47.31
Difficulty-Net [43]	91.92	73.31	86.25	83.76	76.09	73.38	71.76	73.27	35.26	50.47	71.32	58.04
EME-C (Ours)	91.38	85.46	89.08	88.63	77.52	88.10	58.95	74.32	58.27	65.11	69.83	66.08
EME-D (Ours)	**96.31**	**94.77**	**91.42**	**94.24**	**86.90**	92.03	**89.32**	**89.92**	**71.36**	75.68	**73.81**	**73.77**

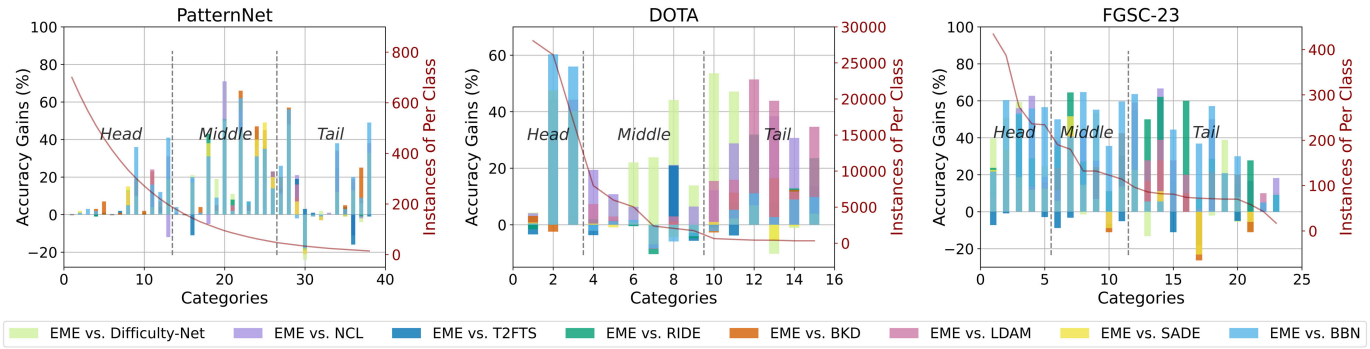


Fig. 5. Performance gain comparison of the proposed EME method and other methods on three datasets, i.e., PatternNet, DOTA, and FGSC-23. The figure depicts the top-1 accuracy gain (%) of EME over others for each category.

B. Implementation Details

We adopt ResNet50 [2] as the feature extractor and a linear layer for both EDors and experts. All input images are resized to 256×256 using bilinear interpolation. For data augmentation, we apply random cropping, random rotation, brightness, and contrast adjustments to the training images. We optimize all networks with SGD, where the learning rate is 0.01, momentum is 0.9, and weight decay is $5e-4$. Cosine annealing is used to adjust the learning rate. In decoupled mode, EDors are trained for 100 epochs while experts are trained for 50 epochs. In coupled mode, EDors and experts are end-to-end trained jointly for 100 epochs. For EDors training, we follow [23] and set $T = 1$ in (4), $m_{\text{id}} = -27$, $m_{\text{ood}} = -3$ in (6), and the weight of energy loss L_e is 0.1. In decision fusion, the energy thresholds γ_1 and γ_2 are searched within the range of $((m_{\text{id}} + m_{\text{ood}})/2 - 5)$ and $((m_{\text{id}} + m_{\text{ood}})/2 + 5)$, respectively. The fusion weights are $\mu_1 = 0.4$ and $\mu_2 = 0.6$.

C. Comparison Experiments

In this section, we compare the proposed EME-C and EME-D with other SOTA methods, including LDAM [39], BBN [20], RIDE [22], BKD [40], SADE [41], T2FTS [7], NCL [42], and Difficulty-Net [43]. EME-C adopts the MEL-C, while EME-D adopts the MEL-D. For fairness, all methods use pre-trained Resnet50 as the backbone.

To provide a comprehensive analysis of the performance of different methods, we not only present the results on the entire dataset but also consider subsets of data: the head data, middle data, and tail data. The head data comprises categories with a large number of samples, the middle data consists of categories with a moderate sample size, and the tail data contains fewer samples per class. Table II displays the comparison results of top-1 accuracy (in percentage) for these datasets. Notably, The imbalance ratio of the PatternNet dataset is 0.02. To provide a clearer demonstration of the advantages offered by our method, we have generated Fig. 5, which depicts the improvements achieved when compared with other methods. This visual representation allows for a more intuitive understanding of the benefits and enhancements provided by our approach. Our experiments have yielded several noteworthy observations:

1) Superior Overall Accuracy: Our method consistently outperforms other approaches in terms of overall accuracy across all three datasets. Specifically, compared to alternative methods, our method achieves enhancements of $4.7\% - 14.8\%$ on PatternNet, $2.6\% - 18.0\%$ on DOTA, and $0.3\% - 36.2\%$ on FGSC.

2) Decoupled Method Outperforms Coupled Method: When comparing the coupled and decoupled methods (i.e., EME-C and EME-D), the decoupled approach exhibits superior performance. This finding suggests that separating the

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TABLE III
PERFORMANCE COMPARISON OF THE PROPOSED EME MODEL AND OTHER METHODS ON THE PATTERNNET DATASET UNDER FOUR IMBALANCE RATIOS OF 0.1, 0.05, 0.02, 0.01

Method	PatternNet ($\alpha = 0.1$)				PatternNet ($\alpha = 0.05$)				PatternNet ($\alpha = 0.02$)				PatternNet ($\alpha = 0.01$)			
	Head	Middle	Tail	All	Head	Middle	Tail	All	Head	Middle	Tail	All	Head	Middle	Tail	All
LDAM [39]	92.38	83.92	88.75	88.34	91.00	76.46	87.33	84.87	90.38	76.54	78.75	81.97	92.62	75.15	72.58	80.32
BKD [40]	91.62	82.38	86.58	86.87	92.54	77.85	85.25	85.21	90.15	72.46	78.25	80.34	88.38	69.31	70.75	76.29
SADE [41]	92.31	83.54	90.25	88.66	91.85	81.85	89.25	87.61	92.38	74.46	82.75	83.21	91.23	70.00	78.42	79.92
T2FTS [7]	92.85	**88.85**	**91.92**	**91.18**	94.31	91.54	**94.83**	93.53	94.08	87.69	88.75	90.21	94.92	90.54	**87.42**	91.05
BBN [20]	89.15	79.08	86.00	84.71	89.15	77.23	83.33	83.24	87.38	74.23	76.50	79.45	87.92	71.54	76.00	78.55
RIDE [22]	91.85	82.54	87.83	87.39	91.23	77.54	87.00	85.21	92.15	75.00	83.25	83.47	91.08	68.15	78.83	79.37
NCL [42]	95.08	86.77	90.33	90.74	92.62	85.69	94.00	90.68	95.00	84.46	89.08	89.53	91.15	82.00	84.42	85.89
Difficulty-Net [43]	92.92	86.31	91.50	90.21	92.77	78.00	89.00	86.53	91.92	73.31	86.25	83.76	90.92	62.69	72.58	75.47
EME-C (Ours)	91.31	73.62	89.75	84.76	94.15	89.08	89.42	90.92	91.38	85.46	89.08	88.63	92.23	86.69	84.83	88.00
EME-D (Ours)	**96.31**	84.15	90.92	90.45	**95.62**	**96.08**	93.58	**95.13**	**96.31**	**94.77**	**91.42**	**94.24**	**96.23**	**92.85**	87.00	**92.16**

influence of different domains proves advantageous in achieving higher accuracy.

3) Head Data Analysis: Focusing solely on the head results, our method shows clear advantages on all three datasets. This outcome indicates that our method effectively addresses the challenges stemming from tail-interference, leading to improved performance.

4) Middle Data Analysis: For the middle data, the performance of our method is affected by the intrinsic properties of the model architecture. We train two discriminators to classify data into head, middle, and tail subsets. This places middle data at the decision boundary, making it more prone to conflicts during fusion. However, our decision fusion mitigates such issues, enabling performance comparable to other methods on middle categories.

5) Tail Data Analysis: Significantly, our method consistently attains superior results when examining the tail data across all datasets, outperforming other methods. This finding suggests that our method effectively mitigates the impact of head-domain dominance on tail categories, resulting in improved accuracy.

6) Performance Under Varying Degrees of Class Imbalance: Table III presents the performance comparison of various methods on the PatternNet dataset across different class imbalance ratios of 0.1, 0.05, 0.02, and 0.01. Overall, our approach achieves superior results. The sole exception is at an imbalance ratio of 0.1, where our model's overall accuracy is marginally lower than T2FTS [7] and NCL [42]. As the degree of imbalance increases further, our method consistently outperforms all other methods. Notably, at ratios of 0.05, 0.02 and 0.01, representing more severely skewed distributions, our approach attains the highest performance. This indicates that our framework can more effectively mitigate the degradation in performance exhibited by alternative approaches as the data distribution shifts toward a longer tail. Significantly, our model maintains stable and SOTA results even under highly skewed conditions, demonstrating strong robustness to class imbalance.

In summary, our method excels in terms of overall accuracy across all subsets of data. It outperforms alternative methods and effectively tackles the challenges posed by both tail-interference and head-domain dominance.

D. Ablation Studies

In this section, we conduct ablation studies to assess the efficiency of CGD and MEL blocks.

1) CGD Block: Let's examine the results in rows ① and ④ of Table IV. Notably, when the CGD block is absent, there is a significant decrease in overall accuracy by 32.7%. The head, middle, and tail categories also experience substantial reductions of 10.9%, 12.7%, and 63.6%, respectively. This stark drop clearly highlights the critical importance of the CGD block within the overall approach. Its presence significantly contributes to the effectiveness of the method, as demonstrated by the remarkable performance difference between the two scenarios. These findings underscore the pivotal role played by the CGD module in improving the accuracy across all categories, thereby emphasizing its significance in the proposed approach. Next, we focus on each component in CGD.

First, let's evaluate the influence of the number of EDors on the results. By comparing lines ② and ③ in Table IV, we can observe the impact of the number of EDors on the results. The findings indicate that employing two EDors leads to an overall accuracy increase of 7.1% compared to using only one EDor. Furthermore, it is evident that the head, middle, and tail sections also experience respective increases of 2.8%, 11.9%, and 4.3%. These results highlight the crucial significance of the collaborative efforts between two EDors. It should be noted that the CGD module and the MEL module are designed to work in tandem. In the scenario where only one EDor is employed (line ②), the data is partitioned into head and tail segments, allowing for the utilization of only two experts within the MEL module.

Additionally, we conducted carefully designed experiments to assess the individual capabilities of different EDors and their collaborative capacities. The results of these experiments are presented in Table V. Upon examining the table, it is evident that the EDors have demonstrated impressive accuracy levels across all three datasets. Notably, the PatternNet dataset exhibited outstanding performance. Specifically, after decision fusion in the CGD module, the PatternNet dataset maintained high classification accuracies of 99.62% and 98.25% for the head and tail classes respectively from both EDors. This remarkable achievement provides a solid foundation for the seamless operation of the subsequent multiexpert module, ensuring robust protection for its overall functionality.

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TABLE IV
ABLATION STUDY OF DIFFERENT BLOCKS ON THE DOTA DATASET

	CGD Block			MEL Block			Head	Middle	Tail	All
	EDor-1	EDor-2	Decision fusion	φ_1	φ_2	φ_3				
①				✓			75.99	79.37	25.73	57.24
②	✓			✓	✓		84.12	80.09	84.26	82.56
③	✓	✓		✓	✓	✓	86.90	92.02	88.53	89.61
④	✓	✓	✓	✓	✓	✓	**86.92**	**92.03**	**89.32**	**89.92**

TABLE V
PERFORMANCE OF EDOR-1 AND EDOR-2 ON DIFFERENT DATASETS

Module		PatternNet	DOTA	FGSC-23
EDor-1	H	99.62	96.68	85.20
	MT	99.68	98.77	81.71
EDor-2	HM	100	98.41	96.08
	T	98.25	94.43	79.72
EDor-1+EDor-2 (CGD)	H	99.62	96.63	81.89
	M	99.69	94.11	80.00
	T	98.25	94.32	77.83

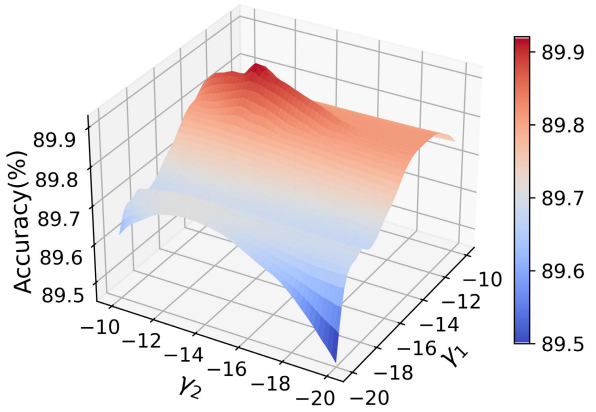


Fig. 6. Influence of energy thresholds γ_1 and γ_2 on the DOTA dataset.

Next, let us shift our attention to the significance of Decision fusion. In Table IV, line ③ represents the outcome achieved by directly connecting two EDors in series, while line ④ demonstrates the result obtained when the fusion strategy is employed to connect the two EDors in parallel. By comparing these two rows, we can observe that the fusion strategy leads to an enhancement of approximately 0.3% in terms of overall accuracy. This highlights the positive impact of decision fusion on improving the performance of the system.

2) MEL Block: Comparing lines ①, ②, and ③ in Table IV reveals the significant impact of the number of experts on the results. Transitioning from one expert to two experts yields a remarkable improvement in overall accuracy by 25.3%. Additionally, the accuracy of the head, middle, and tail data segments experiences notable enhancements of 8.1%, 0.7%, and 58.5%, respectively. Further, increasing the number of experts from two to three leads to an overall accuracy improvement of 7.1%, with corresponding improvements of 2.8%, 11.9%, and 4.3% in the accuracy of the head, middle, and tail sections, respectively. Incorporating more experts consistently

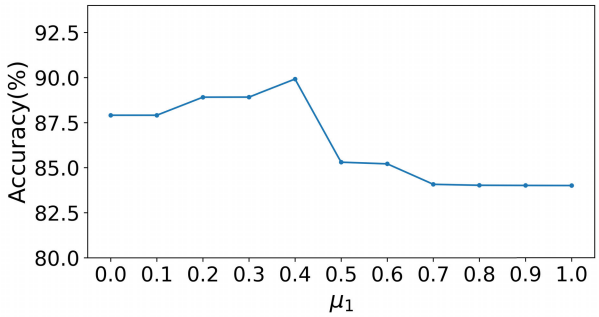


Fig. 7. Influence of weights μ_1 and μ_2 on the DOTA dataset, where $\mu_2 = 1 - \mu_1$.

TABLE VI
INFLUENCE OF DI STRATEGY ON THE DOTA DATASET

Overall Accuracy (%)	
w/o Decision-Interaction	**w/ Decision-Interaction**
31.44	40.81
63.69	67.68
91.61	92.55

contributes to accuracy gains across all data segments, with the tail segment exhibiting the most substantial improvement initially, and the middle segment showing relatively larger gains when increasing from two to three experts.

In our MEL block, we have integrated a DI strategy during the inference stage to assist in completing the classification task. This module plays a critical role in rectifying potential errors generated by the CGD block. To assess the effectiveness of the DI module, we conducted an ablation experiment, and the results are presented in Table VI, which compares three scenarios. The first column displays the results obtained when the DI strategy is not utilized during the inference stage, while the second column presents the outcomes achieved when the DI strategy is employed. The experimental results clearly demonstrate that when the performance of the CGD block is suboptimal, the DI strategy can significantly enhance the overall accuracy of the model. However, it is worth noting that when the CGD module itself achieves high accuracy, the impact of the DI module becomes less pronounced.

E. Influence of Hyperparameters

1) Influence of Energy Thresholds: The energy thresholds γ_1 and γ_2 in (8) and (9) play a crucial role in the decision fusion process. To further elucidate their effects, we conducted an ablation study by systematically varying the threshold values.

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TABLE VII
COMPARISON OF ENERGY-BASED AND LOGITS-BASED MODELS WHEN CLASSIFYING HEAD, MIDDLE, AND TAIL DATA ON THE DOTA DATASET

CGD	Head	Middle	Tail	All
w/o energy	70.91	**93.97**	81.85	84.51
w/ energy	**86.92**	92.03	**89.32**	**89.92**

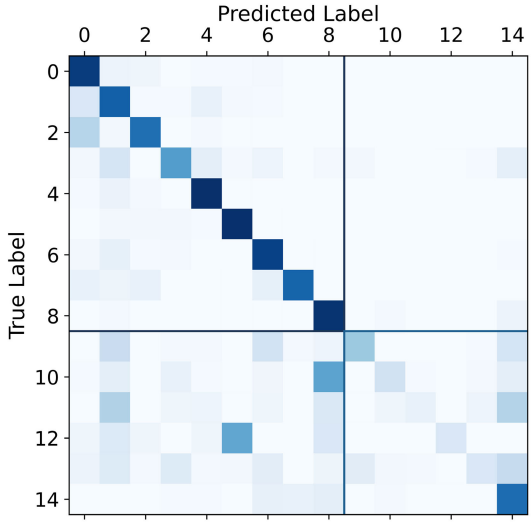


Fig. 8. Confusion matrix of the model trained on the DOTA dataset. Head classes (0–8) have higher recall than tail classes (9–14). Tail classes are more misclassified as heads.

Specifically, γ_1 and γ_2 were each sampled independently within the range $[-20, -10]$ in increments of 0.5, forming different combinations of threshold pairs. The overall accuracy for each configuration is depicted in Fig. 6. As can be observed, as γ_1 increases from -20 to -10, the accuracy initially rises and then decreases, exhibiting a prominent peak. This suggests that γ_1 values near the peak optimize the separation between the ID and OOD regions for EDor-1. In contrast, changes to γ_2 over the test interval elicit a relatively minor fluctuation in accuracy. This implies that EDor-2 is less sensitive to alterations in its threshold. Maximum accuracy was obtained when both γ_1 and γ_2 approached -12. This threshold pairing appears to achieve the best balance between the decisions of the two EDors.

2) Influence of Weights: The weights μ_1 and μ_2 (where $\mu_2 = 1 - \mu_1$) in (10) have a significant impact on the final outcome. To assess the influence of these weights, we conducted experiments and analyzed their sensitivity. The results are shown in Fig. 7. Analysis reveals that the optimal performance is achieved when μ_1 is set to 0.4 and μ_2 is set to 0.6. By assigning a higher weight to E_2, we emphasize the energy contribution from the EDor-2, while still considering the information from the EDor-1 through its lower weight μ_1. This balanced weighting allows the decision fusion module to leverage the complementary strengths of both experts, leading to improved overall performance.

In summary, this study revealed differences in the capabilities of EDor-1 and EDor-2. EDor-1 showed greater sensitivity to threshold changes, while EDor-2 exhibited more stable performance over different test conditions. By exploring optimal weight configurations, we were able to maximize the combined strengths of both EDors.

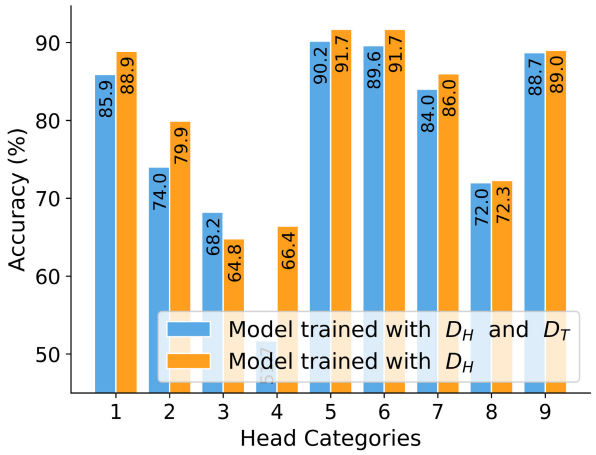


Fig. 9. Accuracy on the head test set for models trained on full dataset vs. head-only. The model trained on head-only dataset has higher accuracy, showing tail-interference.

F. Influence of Energy-Based Model

In the article, EDors utilizes an energy-based model to differentiate between head, middle, and tail data. As an alternative approach, logits can be directly employed for this purpose. To compare the two methods, we conducted an experiment to evaluate the improvement brought by the energy-based approach in comparison to the logits-based approach. The results of the experiment are presented in Table VII. The experimental findings reveal that the energy-based method enhances the overall accuracy by 5.4% when compared to the logits-based method. Moreover, it demonstrates a substantial improvement of 16.0% and 7.5% on the head and tail data, respectively. However, there is a slight decrease in performance observed in the middle data. These results highlight the significance of utilizing the energy model to achieve accurate outcomes.

By optimizing the energy loss, EDors separate positive and negative samples to a greater degree. Specifically, EDors pay more attention to difficult samples located near decision boundaries during training. Logits-based models rely more on cross-entropy loss, which does not prioritize boundary samples to the same extent. When dealing with imbalanced or insufficient training data, logits-based models tend to exhibit reduced performance for minority boundary samples due to weaknesses of cross-entropy optimization. However, EDors demonstrate advantages. As the learning objective is less dependent on sample quantities compared to logits-based models, EDors are more robust against issues like overfitting to majority classes.

G. Experiments for Motivation

As mentioned previously, the long-tailed problem comprises two main factors: Head-dominance and Tail-interference. Our method is specifically designed to mitigate these issues. Therefore, it is essential to examine whether these factors indeed have a negative impact on the results. All the related experiments are conducted on the DOTA dataset.

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First, let us assess the influence of the Head-dominance using the confusion matrix depicted in Fig. 8. The darker shades on the diagonal of the image correspond to higher recall rates. It is evident that the recall rate for the first nine categories (head) is higher compared to the last six categories (tail). Additionally, the head categories are mostly correctly classified without misclassifications into the tail. However, some tail categories are mistakenly assigned to the head, providing clear evidence of the negative impact of Head-dominance on the results.

Next, let us analyze the influence of the Tail-interference factor through Fig. 9. All results reported in the table are evaluated on the head category test data. It is clearly observed that the results obtained when training solely with the head training data outperform the results obtained when training with both the head and tail data. This demonstrates that the presence of tail data significantly interferes with the head data, leading to compromised performance.

These analyses substantiate the detrimental effects of both Head-dominance and Tail-interference on the results, emphasizing the necessity and effectiveness of our method in addressing these challenges.

VI. Conclusion

This article begins by analyzing the reasons behind the poor performance of remote sensing long-tailed data, which are then summarized into two main factors: head-dominance and tail-interference. To address these challenges, the article proposes an EME model. The core concept of this method involves separating the head and tail data to eliminate interference between them. Subsequently, experts are assigned to each separated data segment for classification, leveraging their specialized strengths and knowledge. The effectiveness of our proposed method is demonstrated through excellent performance on three commonly used remote sensing datasets. However, it is important to acknowledge the main limitation of this study, which lies in the subpar performance of the designed CGD module when classifying the middle data. Various factors contribute to this limitation, including the choice of the decision fusion algorithm and the overall design of EDors. As part of future work, we are committed to designing improved algorithms that achieve the goal of noninterference among head, middle, and tail data. Our aim is to efficiently accomplish subsequent tasks while addressing this limitation.

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